Pulmira

An observation-led respiratory digital twin

COPD monitoring research prototype

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The monitoring problem

Respiratory observations arrive from different contexts and clocks.

Sparse or stale readings can hide behind a single dashboard score.

Useful review needs visible coverage and patient-specific history.

Event-linked data must support claims of clinical prediction.

Healthcare workflow

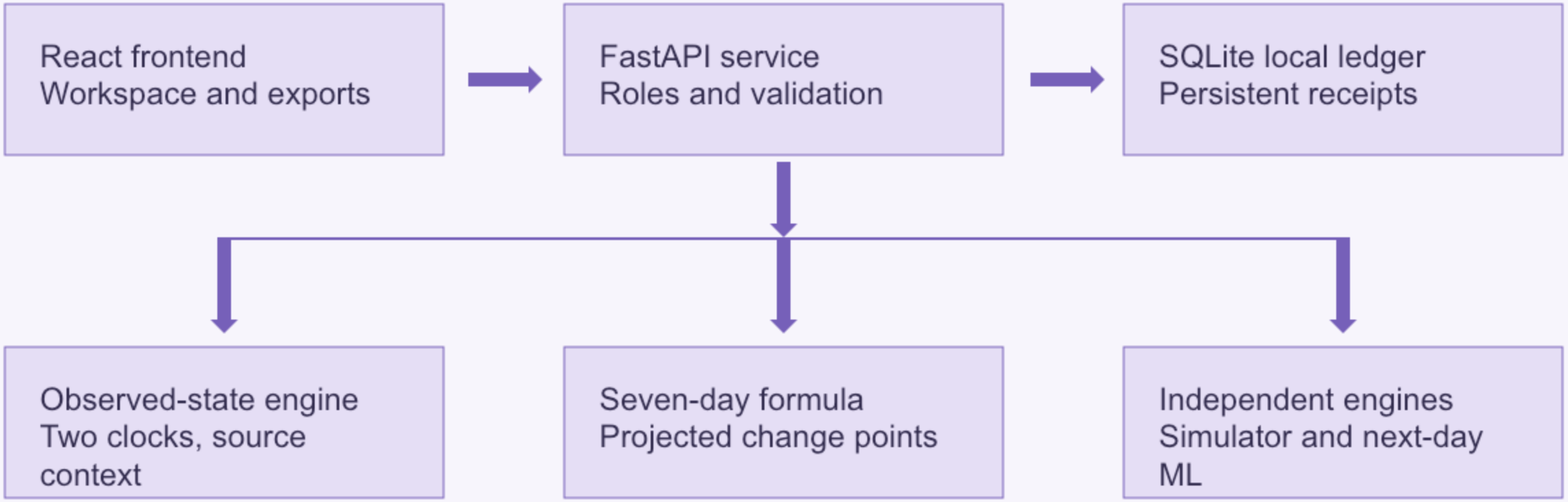
A reviewer examines available observations and their source context.

The workspace shows history, missingness and descriptive changes.

A fictional patient submits symptom check-ins to the local ledger.

Reports preserve the selected historical decision and provenance.

Integrated application architecture



The API calls each engine independently. Public inputs are fictional.

Seven-day formula outlook

Eligible recent history drives robust, bounded trend projections.

Declared weights combine vital changes with recent symptom answers.

The display reports seven future dates with 0-100 change points.

Unknown inputs widen the possible range. Predictive accuracy is unestablished.

Separate next-day ML experiment

Target: next-day simulated recorded SpO2 below 90%.

Histogram gradient boosting uses 102 current, past and static features.

Fictional base profiles: 480 training, 160 validation, 160 held-out test.

The validation-selected threshold stays frozen for test evaluation.

Synthetic results and their limits

95.25% accuracy on 17,442 evaluable synthetic days.

87.23% always-no-warning accuracy.

84.38% precision / 77.11% recall / 84.70% runtime coverage.

Newly low-state recall is only 13.48%. Clinical validity remains unestablished.

Fictional scenario simulation

Sixteen scenarios vary declared engineering assumptions.

Reference and variant runs share keyed exogenous streams.

Hash checks guard the frozen generator and parameter package.

The API returns observed channels and discards private future labels.

Public data and security boundary

The public default uses explicitly fictional monitoring observations.

Real-person observations, local databases and secrets stay outside Git.

Runtime signing secrets are generated locally and ignored by Git.

The dataset source and licence are verified. Artwork rights still need confirmation.

Implemented outcomes

Frontend, backend, Python engines and portable model run together locally.

The workspace supports history, coverage, outlook and review workflows.

PDF summaries and CSV analysis exports complement technical JSON.

Software checks verify behaviour and integrity. Clinical validation remains open.

Validation and submission next steps

Upload the narrated 15-20 minute demo to YouTube as Unlisted.

Retain the dataset attribution and confirm inherited artwork rights.

Acquire event-linked observations with verified timing and surveillance.

Predeclare a patient-separated study, then assess incident recall and calibration.